Lecture 18: Exponentially Affine Models in Discrete Time

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Intro

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- Last lecture: EH failed, forwards forecast returns, Diebold-Li summarized the curve
- Today: build a no-arbitrage model that separates expected rates from term premia
- Goal: move from reduced-form curve dynamics to a structural pricing system
- This is where $\mathbb{P}$, $\mathbb{Q}$, and λ_t start doing real work

Why EH Is Not Enough

- EH was a useful benchmark, but too restrictive
- It ruled out time-varying expected bond returns
- The data did not cooperate (Fama and Bliss 1987; Campbell and Shiller 1991)
- We need a model where the whole curve is priced jointly
- And where expected short rates and term premia are different objects

What This Model Will Deliver

- A state vector X_t
- Dynamics under the physical measure $\mathbb{P}$
- An explicit market price of risk λ_t
- Risk-neutral dynamics under $\mathbb{Q}$
- Bond prices, yields, risk-neutral yields, and term premia

Flight Plan

- Part I: affine setup under $\mathbb{P}$
- Part II: change of measure and why pricing uses $\mathbb{Q}$
- Part III: exponential-affine bond pricing
- Part IV: risk-neutral yields and term premia
- Part V: implementation and estimation with observed factors only

Part I: Setup Under $\mathbb{P}$

State Vector and VAR Under $\mathbb{P}$

- Let $X_t \in \mathbb{R}^K$ collect the term-structure factors
- In this lecture, X_t is *observed*
- Think: first few PCs of yields, or a small set of yields

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim N(0, I_K)$$

- $\mathbb{P}$ is the actual law of motion in the data

Affine Short Rate

$$y_t^{(1)} = \delta_0 + \delta_1^\top X_t$$

- Same notation as last lecture
- The one-period yield is the short rate
- The short end of the curve moves with the state
- Once X_t is given, the short rate is pinned down

SDF and Market Price of Risk

$$m_{t+1} = -y_t^{(1)} - \frac{1}{2}\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}$$

$$\lambda_t = \lambda_0 + \lambda_1 X_t$$

- The SDF comes first: it is the pricing object
- λ_t is the market price of risk embedded inside it
- If λ_t changes with X_t, bond premia change with the state
- No Λ_1: we use λ_1

Economic Interpretation

- $\mathbb{P}$: how the state actually evolves
- m_{t+1}: how payoffs are discounted
- λ_t: how much compensation investors require for bond risk
- If λ_t varies with X_t, term premia vary with X_t
- That is already something EH could not do

Questions?

Part II: Change of Measure

One-Step Density Ratio

Define

$$\left.\frac{d\mathbb{Q}}{d\mathbb{P}}\right|_{t+1} = \exp\left(-\frac{1}{2}\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}\right)$$

- This is the one-step Radon-Nikodym derivative
- It tilts probabilities state by state
- We now need to check that this is a valid density
- Notice that it is a positive random variable

Quick Recall: Gaussian MGF

If $\varepsilon_{t+1} \sim N(0, I_K)$ under $\mathbb{P}$, then for any a_t known at time t,

$$\mathbb{E}_t^{\mathbb{P}}\left[e^{a_t^\top \varepsilon_{t+1}}\right] = \exp\left(\frac{1}{2} a_t^\top a_t\right)$$

- We will use this twice
- First to show the density integrates to one
- Then to prove the shifted shock is standard normal under $\mathbb{Q}$

It Integrates to One

Conditioning on time t,

$$\mathbb{E}_t^{\mathbb{P}}\left[\frac{d\mathbb{Q}}{d\mathbb{P}}\bigg|_{t+1}\right] = e^{-\frac{1}{2}\lambda_t^\top \lambda_t}\mathbb{E}_t^{\mathbb{P}}\left[e^{-\lambda_t^\top \varepsilon_{t+1}}\right]$$

Applying the Gaussian mgf with $a_t = -\lambda_t$,

$$\mathbb{E}_t^{\mathbb{P}}\left[e^{-\lambda_t^\top \varepsilon_{t+1}}\right] = \exp\left(\frac{1}{2}\lambda_t^\top \lambda_t\right)$$

so

$$\mathbb{E}_t^{\mathbb{P}}\left[\frac{d\mathbb{Q}}{d\mathbb{P}}\bigg|_{t+1}\right] = 1$$

The Shifted Shock

Define

$$\varepsilon_{t+1}^{\mathbb{Q}} \equiv \varepsilon_{t+1} + \lambda_t$$

- Under $\mathbb{Q}$, this will become the convenient innovation
- Same state variables, different conditional law
- We now prove explicitly that $\varepsilon_{t+1}^{\mathbb{Q}} \sim N(0, I_K)$ under $\mathbb{Q}$

Mgf Proof Under $\mathbb{Q}$

For any $u \in \mathbb{R}^K$,

$$
\begin{aligned}
\mathbb{E}_t^{\mathbb{Q}}\left[e^{u^\top \varepsilon_{t+1}^{\mathbb{Q}}}\right] &= \mathbb{E}_t^{\mathbb{P}}\left[\frac{d\mathbb{Q}}{d\mathbb{P}}\bigg|_{t+1} e^{u^\top(\varepsilon_{t+1}+\lambda_t)}\right] \\
&= \exp\left(u^\top \lambda_t - \frac{1}{2}\lambda_t^\top \lambda_t\right) \mathbb{E}_t^{\mathbb{P}}\left[e^{(u-\lambda_t)^\top \varepsilon_{t+1}}\right] \\
&= \exp\left(u^\top \lambda_t - \frac{1}{2}\lambda_t^\top \lambda_t + \frac{1}{2}(u-\lambda_t)^\top (u-\lambda_t)\right) \\
&= \exp\left(\frac{1}{2}u^\top u\right)
\end{aligned}
$$

- This is the mgf of $N(0, I_K)$
- So $\varepsilon_{t+1}^{\mathbb{Q}} \sim N(0, I_K)$ under $\mathbb{Q}$

$\mathbb{Q}$-Dynamics of the State

Start from the $\mathbb{P}$-VAR and replace $\varepsilon_{t+1} = \varepsilon^{\mathbb{Q}}_{t+1} - \lambda_t$:

$$X_{t+1} = \mu + \Phi X_t + \Sigma(\varepsilon^{\mathbb{Q}}_{t+1} - \lambda_t)$$

Using $\lambda_t = \lambda_0 + \lambda_1 X_t$,

$$X_{t+1} = \underbrace{(\mu - \Sigma\lambda_0)}_{\equiv \mu^{\mathbb{Q}}} + \underbrace{(\Phi - \Sigma\lambda_1)}_{\equiv \Phi^{\mathbb{Q}}} X_t + \Sigma\varepsilon^{\mathbb{Q}}_{t+1}$$

Pricing Uses $\mathbb{Q}$, Not $\mathbb{P}$

$$P_t^{(n+1)} = \mathbb{E}_t^{\mathbb{P}}[M_{t+1} P_{t+1}^{(n)}] = e^{-y_t^{(1)}} \mathbb{E}_t^{\mathbb{Q}}[P_{t+1}^{(n)}]$$

- There is a state evolution under both $\mathbb{P}$ and $\mathbb{Q}$
- But the bond pricing recursion uses $\mathbb{Q}$
- $\mathbb{P}$ matters for forecasts and expected excess returns
- $\mathbb{Q}$ matters for prices
- This is not “just different initial conditions”: the dynamics themselves change

Questions?

Part III: Bond Pricing

Pricing Equation

For a zero-coupon bond,

$$P_t^{(n+1)} = \mathbb{E}_t^{\mathbb{P}}[M_{t+1} P_{t+1}^{(n)}] = e^{-y_t^{(1)}} \mathbb{E}_t^{\mathbb{Q}}[P_{t+1}^{(n)}]$$

- Last lecture gave the bond math
- Now we impose no arbitrage
- Once we know the short rate and the $\mathbb{Q}$-dynamics, all bond prices are pinned down

Exponential-Affine Guess

Guess (Duffie and Kan 1996)

$$P_t^{(n)} = \exp(A_n + B_n^\top X_t)$$

with

$$A_0 = 0, \qquad B_0 = 0$$

- $P_t^{(0)} = 1$ is the base case
- The question is whether this functional form closes under the recursion

Plug the Guess Into the Pricing Equation

Use

$$y_t^{(1)} = \delta_0 + \delta_1^\top X_t$$

and

$$X_{t+1} = \mu^{\mathbb{Q}} + \Phi^{\mathbb{Q}} X_t + \Sigma \varepsilon_{t+1}^{\mathbb{Q}}$$

Then

$$P_t^{(n+1)} = e^{-\delta_0 - \delta_1^\top X_t} \mathbb{E}_t^{\mathbb{Q}} \left[\exp \left(A_n + B_n^\top X_{t+1} \right) \right]$$

Compute the Gaussian Expectation

Substitute for X_{t+1} and collect what is known at time t:

$$P_t^{(n+1)} = e^{-\delta_0 - \delta_1^\top X_t} \mathbb{E}_t^{\mathbb{Q}} \left[\exp\left(A_n + B_n^\top (\mu^{\mathbb{Q}} + \Phi^{\mathbb{Q}} X_t + \Sigma \varepsilon_{t+1}^{\mathbb{Q}}) \right) \right]$$

$$P_t^{(n+1)} = \exp\left(A_n - \delta_0 + B_n^\top \mu^{\mathbb{Q}} + [(\Phi^{\mathbb{Q}})^\top B_n - \delta_1]^\top X_t\right) \mathbb{E}_t^{\mathbb{Q}} \left[e^{B_n^\top \Sigma \varepsilon_{t+1}^{\mathbb{Q}}} \right]$$

Because $\varepsilon_{t+1}^{\mathbb{Q}} \sim N(0, I_K)$,

$$\mathbb{E}_t^{\mathbb{Q}} \left[e^{B_n^\top \Sigma \varepsilon_{t+1}^{\mathbb{Q}}} \right] = \exp\left(\frac{1}{2} B_n^\top \Sigma \Sigma^\top B_n \right)$$

Affine Recursions

Matching coefficients gives

$$A_{n+1} = A_n + B_n^\top \mu^{\mathbb{Q}} + \frac{1}{2} B_n^\top \Sigma\Sigma^\top B_n - \delta_0$$

$$B_{n+1} = (\Phi^{\mathbb{Q}})^\top B_n - \delta_1$$

- The recursion closes
- This is why the model is called exponentially affine

Yields Are Affine Too

Using

$$y_t^{(n)} = -\frac{1}{n} \log P_t^{(n)}$$

we get

$$y_t^{(n)} = -\frac{1}{n}(A_n + B_n^\top X_t)$$

- Each maturity loads linearly on the same state vector
- The loadings vary with n through A_n and B_n
- The whole curve is one joint pricing object

Connection Back to Lecture 17

- Diebold-Li also used a small number of factors to describe the curve (Diebold and Li 2006)
- But there the loadings were chosen for fit and forecasting
- Here the loadings come from no-arbitrage restrictions (Duffee 2002; Ang and Piazzesi 2003)
- Most important: the model gives a disciplined $\mathbb{P}/\mathbb{Q}$ split

Questions?

Part IV: EH Component and Term Premia

The EH Component Inside the Model

Lecture 17 defined the expectations-hypothesis piece as

$$EH_t^{(n)} \equiv \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^{\mathbb{P}} \left[y_{t+j}^{(1)} \right]$$

- This is the average expected future short rate
- It is a $\mathbb{P}$-expectations object
- It is *not* computed from the bond-pricing recursion
- Here we show how to compute it once the affine model is estimated

Forecast the State Under $\mathbb{P}$

Start from

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}, \qquad \mathbb{E}_t^{\mathbb{P}}[\varepsilon_{t+1}] = 0$$

Then

$$\mathbb{E}_t^{\mathbb{P}}[X_{t+1}] = \mu + \Phi X_t$$

and, recursively,

$$\mathbb{E}_t^{\mathbb{P}}[X_{t+j}] = \sum_{i=0}^{j-1} \Phi^i \mu + \Phi^j X_t$$

Forecast Future Short Rates

Using

$$y_t^{(1)} = \delta_0 + \delta_1^\top X_t$$

we get

$$\mathbb{E}_t^{\mathbb{P}}\left[y_{t+j}^{(1)}\right] = \delta_0 + \delta_1^\top \mathbb{E}_t^{\mathbb{P}}[X_{t+j}]$$

So once $(\mu, \Phi, \delta_0, \delta_1)$ and X_t are known,

$$\mathbb{E}_t^{\mathbb{P}}\left[y_{t+j}^{(1)}\right] = \delta_0 + \delta_1^\top \left(\sum_{i=0}^{j-1} \Phi^i \mu + \Phi^j X_t\right)$$

Compute the EH Component

Now average those expected short rates:

$$EH_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^{\mathbb{P}} \left[y_{t+j}^{(1)} \right]$$

Algorithm:

1. Start from the estimated state X_t
2. Forecast $X_{t+1}, \dots, X_{t+n-1}$ under the (model-implied) $\mathbb{P}$
3. Map each state forecast into an expected short rate
4. Average across the n horizons

Observed, Fitted, and EH Component

- Observed yield: $y_t^{(n)}$
- Fitted/model-implied yield: $\hat{y}_t^{(n)}$
- EH component: $EH_t^{(n)}$
- Term premium: $TP_t^{(n)}$

Observed, Fitted, and EH Component

- Observed yield: $y_t^{(n)}$
- Fitted/model-implied yield: $\hat{y}_t^{(n)}$
- EH component: $EH_t^{(n)}$
- Term premium: $TP_t^{(n)}$
- In an exact-fit model, $y_t^{(n)}$ and $\hat{y}_t^{(n)}$ are the same
- In an estimated model with measurement error, they need not be
- But $EH_t^{(n)}$ is different from both: it uses only $\mathbb{P}$-expected short rates

Term Premia Relative to the EH Component

In the spirit of Lecture 17, define

$$TP_t^{(n)} = y_t^{(n)} - EH_t^{(n)}$$

or, in an inexact-fit model,

$$TP_t^{(n)} = \hat{y}_t^{(n)} - EH_t^{(n)}$$

- This is the part of the yield not attributed to expected future short rates
- If $TP_t^{(n)}$ rises, yields rise relative to the EH component
- If $TP_t^{(n)}$ falls, yields move closer to the EH component
- This creates a gap between *observed* yields and *expected* short rates

Why λ_t Still Matters

- The EH component uses $\mathbb{P}$ forecasts of future short rates
- The fitted yield comes from the no-arbitrage pricing recursion
- The wedge between $\mathbb{P}$ and $\mathbb{Q}$ is driven by $\lambda_t = \lambda_0 + \lambda_1 X_t$
- So λ_t is still the key force behind time-varying term premia

Questions?

Part V: Implementation and Estimation

Inputs After Estimation

Suppose we have estimated

$$\mu,\ \Phi,\ \Sigma,\ \delta_0,\ \delta_1,\ \lambda_0,\ \lambda_1$$

Then compute

$$\mu^{\mathbb{Q}} = \mu - \Sigma\lambda_0, \qquad \Phi^{\mathbb{Q}} = \Phi - \Sigma\lambda_1$$

- The fitted yield curve still uses the pricing recursion with $\mu^{\mathbb{Q}}, \Phi^{\mathbb{Q}}$
- The EH component uses the $\mathbb{P}$-dynamics (μ, Φ)
- After estimation, both objects are available

EH Component Algorithm

For each horizon j,

$$\mathbb{E}_t^{\mathbb{P}}[X_{t+j}] = \sum_{i=0}^{j-1} \Phi^i \mu + \Phi^j X_t$$

Then compute the expected future short rate:

$$\mathbb{E}_t^{\mathbb{P}}\left[y_{t+j}^{(1)}\right] = \delta_0 + \delta_1^\top \mathbb{E}_t^{\mathbb{P}}[X_{t+j}]$$

Finally,

$$EH_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^{\mathbb{P}}\left[y_{t+j}^{(1)}\right]$$

From Fitted Yields to Term Premia

1. Use the no-arbitrage recursion to compute fitted yields $\hat{y}_t^{(n)}$
2. Use the $\mathbb{P}$-VAR and the short-rate equation to compute $EH_t^{(n)}$
3. Define term premia as

$$TP_t^{(n)} = \hat{y}_t^{(n)} - EH_t^{(n)}$$

In an exact-fit model, this is the same as

$$TP_t^{(n)} = y_t^{(n)} - EH_t^{(n)}$$

Questions?

Estimation Route: Observed Factors Only

1. Choose observed factors X_t
2. Estimate the VAR under $\mathbb{P}$ by OLS
3. Estimate the short-rate equation
4. Choose λ_0, λ_1 to match the yield curve
5. Recover fitted yields, the EH component, and term premia

See also Joslin et al. (2011) and Adrian et al. (2013).

Matching the Yield Curve

A simple objective is

$$\min_{\lambda_0,\lambda_1} \sum_t \sum_{n\in\mathcal{M}} \left(y_t^{(n)} - \widehat{y}_t^{(n)}\right)^2$$

- $\mathcal{M}$: maturities used in estimation
- The $\mathbb{P}$-VAR alone is not enough
- The cross section disciplines the pricing wedge

What the Estimated Model Delivers

- Fitted yield curves
- Risk-neutral yield curves
- Term-premium estimates by maturity and over time
- A clean separation between expected-rate news and premium news
- A bridge from term-structure data to macro and finance questions

Questions?

Taking Stock

- Lecture 17 told us the curve reflects both expected rates and premia
- Lecture 18 gave a no-arbitrage model that makes that separation explicit
- $\mathbb{P}$ describes the data; $\mathbb{Q}$ prices bonds
- λ_t is the wedge between the two
- Exponential-affine recursions turn that wedge into bond prices, yields, and term premia

Readings and References i

- Term-structure overview: Gürkaynak and Wright (2012)
- Evidence against EH: Fama and Bliss (1987); Campbell and Shiller (1991)
- Reduced-form benchmark: Diebold and Li (2006)
- Affine term-structure foundations: Duffie and Kan (1996); Duffee (2002); Ang and Piazzesi (2003)
- Observable-factor estimation: Joslin et al. (2011); Adrian et al. (2013)

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